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Mixed finite element analysis of a non-linear three-fields Stokes model

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In this article, a mixed finite element analysis of the non-linear Stokes problem with monotone constitutive laws is considered. We construct a new three-field model for incompressible fluids where the velocity $\mathbf{u}$, the non-linear stress tensor $\sigma = \Psi(|\nabla\mathbf{u}|)\nabla\mathbf{u}$ and the pressure p are the most relevant unknowns. We give existence and unicity results for the continuous problem and its approximation. Stable and optimal error estimates under minimal regularity assumptions are derived and numerical results are presented.

1. Introduction

It is well known that the flow of materials with low numbers of molecules, such as water and air, can be modelled by Navier–Stokes equations. The flows of materials with macromolecules, such as biological fluids, lubricators, paint and high polymer solutions, are usually described by more complicated constitutive equations. In fact, considerations of the motion of rheologically complex fluids lead to the system of Stokes or Navier–Stokes equations with non-linear constitutive laws. In this paper we study the non-linear model for the Stokes equations, i.e. it is assumed that the viscosity of the medium is a non-linear monotone function of the gradient of the velocity. More precisely, the problem that we consider is the following generalized Stokes problem:

$$\begin{cases} \text{Find } (\mathbf{u}, p) \text{ such that} \\ -\nabla\cdot(\Psi(|\nabla\mathbf{u}|)\nabla\mathbf{u}) + \nabla p = \mathbf{f} \quad \text{in } \Omega \\ \nabla\cdot\mathbf{u} = 0 \quad \text{in } \Omega \\ \mathbf{u} = 0 \quad \text{on } \Gamma \end{cases} \tag{1.1}$$

where Ω is a bounded open subset of $\mathbb{R}^2$ with a Lipschitz continuous boundary Γ, $\mathbf{u}$ denotes the velocity, p is the pressure, $\mathbf{f}$ is the external forces, and $\Psi : \mathbb{R}^+ \longrightarrow \mathbb{R}^+$ is a continuous and non-linear kinematic viscosity function of the fluid. Here, and throughout the paper, we use the following notations:

- $\nabla\cdot\mathbf{u}$ is the divergence of $\mathbf{u}$;

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- $\nabla \mathbf{u}$ is the tensor gradient of $\mathbf{u}$;
- $|\nabla \mathbf{u}| = \left(\sum_{i,j=1}^{2} \left| \frac{\partial u_i}{\partial x_j} \right|^2 \right)^{\frac{1}{2}}$;
- $\sigma = (\sigma_{ij}), \tau = (\tau_{ij}), \quad \sigma : \tau = \sum_{i \cdot j=1}^{2} \sigma_{ij} \tau_{ij}, \quad |\sigma|^2 = \sigma : \sigma.$

Three classical laws for Ψ are as follows:

- *Ladyzhenskaya law* (cf. Ladyzhenskaya (1970))

$$\Psi(|\nabla \mathbf{u}|) = \nu_0 + \nu_1 |\nabla \mathbf{u}|^{\beta-2}, \quad \nu_0 \geqslant 0, \quad \nu_1 > 0, \quad \beta > 1 \tag{1.2}$$

for fluids with large stresses.

- *Odqvist–Norton power law*

$$\Psi(|d(\mathbf{u})|) = \nu_0 |d(\mathbf{u})|^{\beta-2}, \quad \nu_0 > 0, \quad \beta > 1 \tag{1.3}$$

where $d(\mathbf{u})$ denotes the deformation tensor, i.e.

$$d(\mathbf{u}) = \tfrac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^t).$$

- *Carreau's law*

$$\Psi(|d(\mathbf{u})|) = \nu_0 (1 + |d(\mathbf{u})|^2)^{\frac{\beta-2}{2}}, \quad \nu_0 > 0, \quad \beta \geqslant 1 \tag{1.4}$$

for visco-plastic flows and creeping flow of metal.

These laws occur in many mathematical models of physical process of polymers, or of visco-elastic or visco-plastic fluids, or of fluids with large stresses (see Ladyzhenskaya (1970), Byron & Amstrong (1987), Loula & Guerreiro (1990)). In the mixed formulation that we will consider, we shall use the inversion of the relation $\sigma = \Psi(|\nabla \mathbf{u}|)\nabla \mathbf{u}$ to obtain $\nabla \mathbf{u}$ as a function of σ. For this purpose, let $g = (g_{ij})_{1 \leqslant i,j \leqslant 2}$ where

$$g_{ij} : I\!R^4 \longrightarrow I\!R, \quad g_{ij}(\tau) = \Psi(|\tau|)\tau_{ij}$$

is a twice continuously differential with bounded derivative through the second order. We shall assume that the function g satisfies the following hypothesis:

$$\forall \tau \in I\!R^4, \tau \neq 0, \forall \theta \in I\!R^4, \quad \sum_{i,j,k,l=1}^{2} \frac{\partial g_{ij}}{\partial \tau_{kl}} \theta_{ij} \theta_{kl} \geqslant C_0 \|\theta\|^2_{I\!R^4}. \tag{1.5}$$

Owing to the implicit function theorem, the ellipticity condition (1.5) guarantees that the relation $\sigma = \Psi(|\nabla \mathbf{u}|)\nabla \mathbf{u}$ can be inverted to obtain $\nabla \mathbf{u}$ as a function of σ, say

$$\nabla \mathbf{u} = \Phi(\sigma). \tag{1.6}$$

For example, we have the following relation for the power law.

PROPOSITION 1 Let $\beta > 1$. Then

$$\sigma = |\nabla \mathbf{u}|^{\beta-2}\nabla \mathbf{u} \iff \nabla \mathbf{u} = \Phi_\alpha(\sigma) := |\sigma|^{\alpha-2}\sigma$$

where $\frac{1}{\alpha} + \frac{1}{\beta} = 1$.

In this paper we analyse a mixed finite formulation of the problem (1.1) for the Ladyzhenskaya law given by (1.2) with $\nu_0 = 0$ and $\nu_1 = 1$, i.e. the Newtonian part of the viscosity set to zero (for ease of exposition). However, we note that, for the mixed finite element method that we consider in this paper (see Section 3), it is very difficult to work with the deformation tensor $d(\mathbf{u})$. The difficulty lies essentially in the symmetry of this tensor. One way to overcome this difficulty is to relax the symmetry of $d(\mathbf{u})$ by a Lagrange multiplier. This problem will be addressed elsewhere. In this mixed formulation, the flux variable

$$\sigma = \Phi_\alpha^{-1}(\nabla \mathbf{u}) \tag{1.7}$$

is introduced as a new unknown in addition to the velocity and the pressure of the fluid. Combining (1.1) and (1.7) the following mixed problem can be derived:

$$\begin{cases} \text{Find}(\sigma, p, \mathbf{u}) \text{ such that} \\ \qquad -\nabla \cdot (\sigma - pI) = \mathbf{f} \qquad \text{in } \Omega \\ \qquad \Phi_\alpha(\sigma) = \nabla \mathbf{u} \quad \text{in } \Omega \\ \qquad \nabla \cdot \mathbf{u} = 0 \qquad \text{in } \Omega \\ \qquad \mathbf{u} = 0 \qquad \text{on } \Gamma \end{cases} \tag{1.8}$$

where $I = (\delta)_{ij}$ denotes the identity matrix. It should be noted that the finite element approximation for this problem, with the deformation tensor instead, has been proposed and analysed by several authors (see Baranger & Najib (1990), Loula & Guerreiro (1990), Baranger *et al.* (1993), Barrett & Liu (1993, 1994), Lei & Tian-Xiao (1998)). One of these works which is close to the one here is Baranger *et al.* (1993). However, their formulation is different from the one considered in this paper. The advantages of our formulation are two-fold: by introducing the tensor $\sigma - pI$, we have the local (i.e. at element level) conservation of the momentum plus the local conservation of the mass; owing to the introduction of Lagrange multipliers (see Section 4), the linear algebraic system which needs to be solved involves only these Lagrange multipliers and the pressure. Moreover, in the case of the lowest degree, the Lagrange multipliers are constants on each edge of the triangulation and the pressure is constant on each triangle. An outline of the paper is as follows. In Section 2 a general mixed variational formulation of the problem (1.8) is presented. Existence and unicity results are then discussed. The mixed finite element approximation is introduced and analysed in Section 3. Stable and optimal error estimates under minimal regularity assumptions are then derived. Finally in Section 4 some numerical results are presented.

2. Abstract non-linear mixed variational formulation in Banach spaces

In this section we recall from Scheurer (1977) some results for a general class of non-linear problems with constraints such as the Navier–Stokes equation in the $(\sigma, p, \mathbf{u})$ formulation.

2.1 *Setting of the problem*

Let X and V be two (real) Banach space reflexives with norms $\|\cdot\|_X$ and $\|\cdot\|_V$, respectively. Let X' and V' be their corresponding dual spaces and let $\|\cdot\|_{X'}$ and $\|\cdot\|_{V'}$ denote their dual norms. We denote by $\langle\cdot,\cdot\rangle$ the duality between X and X' or V and V'. Let $\mathcal{A} : X \longrightarrow X'$ be a non-linear operator and $\mathcal{B} : X \longrightarrow V'$ a linear operator. Let $\mathcal{B}' \in \mathcal{L}(V, X')$ be the dual operator of $\mathcal{B}$ defined by

$$\langle \mathcal{B}'\mathbf{v}, \tau\rangle = \langle \mathbf{v}, \mathcal{B}\tau\rangle, \forall \tau \in X, \forall \mathbf{v} \in V.$$

We consider the following problem. Given $\mathbf{f} \in V'$ and $g \in X'$,

$$\begin{cases} \text{find } \sigma \in X \quad \text{and } \mathbf{u} \in V \text{ such that} \\ \qquad\qquad \mathcal{A}\sigma + B'\mathbf{u} = g \quad \text{in } X', \\ \qquad\qquad \mathcal{B}\sigma = \mathbf{f} \quad \text{in } V'. \end{cases} \tag{2.1}$$

Now let

$$Z(\mathbf{f}) = \{\tau \in X, \mathcal{B}\tau = \mathbf{f}\}.$$

The continuity of $\mathcal{B}$ implies that $Z(\mathbf{f})$ is a closed subspace of X.

We associate with (2.1) the following problem. Given $\mathbf{f} \in V'$ and $g \in X'$

$$\begin{cases} \text{find } \sigma \in Z(\mathbf{f}), \text{ such that} \\ \langle \mathcal{A}\sigma, \tau\rangle = \langle g, \tau\rangle \quad \forall \tau \in Z(0) \end{cases} \tag{2.2}$$

where $Z(0) = \{\tau \in X, \mathcal{B}\tau = 0\} := \text{Ker}(\mathcal{B})$.

We first quote the following abstract theorem concerning the existence and the uniqueness of solutions of mixed problems in Banach spaces (see Scheurer (1977)).

THEOREM 1 We assume that the following hypotheses hold:

(A1) $\mathcal{A}$ is hemicontinuous from X to X'.

(A2) There exists a constant $C_1 > 0$, such that, for $\alpha \geqslant 2$,

$$\langle \mathcal{A}\sigma - \mathcal{A}\tau, \sigma - \tau\rangle \geqslant C_1\|\sigma - \tau\|_X^{\alpha}, \forall \sigma, \tau \in Z(\mathbf{f}).$$

(A3) There exists a constant $C_2 > 0$, such that, for $\alpha \geqslant 2$,

$$\|\mathcal{A}\sigma - \mathcal{A}\tau\|_{X'} \leqslant C_2(\|\sigma\|_X + \|\tau\|_X)^{\alpha-2}\|\sigma - \tau\|_X, \forall \sigma, \tau \in X.$$

(B1) There exists a constant $C_3 > 0$ such that

$$\sup_{\tau \in X} \frac{\langle \mathcal{B}\tau, \mathbf{v}\rangle}{\|\tau\|_X} \geqslant C_3\|\mathbf{v}\|_V, \quad \forall \mathbf{v} \in V.$$

Let $(\mathbf{f}, g) \in V' \times X'$. Then the problem (2.1) has at least one solution $(\sigma, \mathbf{u}) \in X \times V$ which satisfies the following *a priori* estimations:

$$\|\sigma\|_X \leqslant \frac{1}{C_1}\left(\|g\| + C_2\left(\frac{\|\mathbf{f}\|}{C_3}\right)^{\alpha-1}\right)^{\frac{1}{\alpha-1}} + \frac{\|\mathbf{f}\|}{C_3} := M, \tag{2.3}$$

$$\|\mathbf{u}\|_V \leqslant \frac{1}{C_3}(\|g\| + C_2 M^{\alpha-1}). \tag{2.4}$$

This result is a direct consequence of Proposition 2.3 in Scheurer (1977). However, in this Proposition 2.3, it is supposed that Ker($\mathcal{B}$) has a complement supplementary set. As pointed out by one anonymous referee, this hypothesis is not necessary. In fact, there is an error in the proof of Lemma 1.3 of Scheurer (1977) on which the proof of Proposition 2.3 is based. We report, in the Annex to this paper, a correct proof given by this anonymous referee.

REMARK 1 The approach used in Baranger *et al.* (1993) to prove the existence of the continuous problem can also be applied here. However, we have opted for another approach, by using the results of Scheurer (1977), which can be used for more general situations. Further, as pointed out by one anonymous referee, it gives us the opportunity to report a corrected proof for some of the results of Scheurer (1977) (see Annex to this paper).

2.2 *Application to the* $(\sigma, p, \mathbf{u})$ *Stokes equations*

We will fit our non-linear Stokes problem (1.8) into the above abstract framework. We assume that Ω is an open bounded polyhedron with a Lipshitz continuous boundary Γ. Let

$$L^\alpha(\Omega) = \left\{q \mid q \text{ measurable }, \int_\Omega |q(x)|^\alpha \,\mathrm{d}x < \infty\right\}, \quad 1 \leqslant \alpha < \infty$$

be the Banach space of functions with absolute values α-integrable in Ω, with the norm

$$\|q\|_{0,\alpha,\Omega} = \left(\int_\Omega |q(x)|^\alpha \,\mathrm{d}x\right)^{\frac{1}{\alpha}}, \quad \forall q \in L^\alpha(\Omega)$$

with its dual space $L^\beta(\Omega)$, with the duality pairing between $L^\alpha(\Omega)$ and $L^\beta(\Omega)$ denoted by $(\cdot, \cdot)$, and define

$$(p, q) = \int_\Omega pq \,\mathrm{d}x, \ \forall p \in L^\alpha(\Omega), \forall q \in L^\beta(\Omega), \ \text{ with } \frac{1}{\alpha} + \frac{1}{\beta} = 1.$$

$$L^\alpha_0(\Omega) = \{q \mid q \in L^\alpha(\Omega), \quad (q, 1) = 0\}.$$

For $1 \leqslant \alpha < \infty$, let

$$W^{1,\alpha}(\Omega) = \{v \in L^\alpha(\Omega), \quad D^k v \in L^\alpha(\Omega), \quad \forall k, \ |k| \leqslant 1\}$$

denote the usual Sobolev space of order one endowed with the norm

$$\|v\|_{1,\alpha,\Omega} = \left(\sum_{|k|\leqslant 1} \|D^k v\|_{0,\alpha,\Omega}\right)^{\frac{1}{\alpha}}.$$

$W_0^{1,\alpha}(\Omega)$ stands for the subspace of $W^{1,\alpha}(\Omega)$ consisting of functions v such that $v = 0$ on Γ. We shall need the space

$$H^\alpha(\mathrm{div},\,\Omega) = \{\mathbf{v} \in (L^\alpha(\Omega))^2, \quad \nabla \cdot \mathbf{v} \in L^\alpha(\Omega)\}$$

with the norm

$$\|\mathbf{v}\|_{H^\alpha(\mathrm{div},\Omega)} = (\|\mathbf{v}\|_{0,\alpha}^\alpha + \|\nabla \cdot \mathbf{v}\|_{0,\alpha}^\alpha)^{\frac{1}{\alpha}}.$$

For $1 < \beta \leqslant 2$, we let $\alpha \geqslant 2$ defined by $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. In order to derive the mixed variational formulation of problem (1.8), we define the Banach space:

$$X = \{(\tau, q) \in (L^\alpha(\Omega))^4 \times L_0^\alpha(\Omega), \quad \nabla \cdot (\tau - qI) \in (L^\alpha(\Omega))^2\}$$

with the norm

$$\|(\tau, q)\|_X = \left(\|\tau\|_{0,\alpha}^\alpha + \|\nabla \cdot (\tau - qI)\|_{0,\alpha}^\alpha + \|q\|_{0,\alpha}^\alpha\right)^{\frac{1}{\alpha}}, \quad \alpha \geqslant 2$$

and the closed subspaces

$$\begin{aligned} Z(\mathbf{f}) &= \{(\tau, q)| \quad (\tau, q) \in X, \quad -\nabla \cdot (\tau - qI) = \mathbf{f}\}, \\ Z(0) &= \{(\tau, q)| \quad (\tau, q) \in X, \quad \nabla \cdot (\tau - qI) = 0\}, \\ V &= (L^\beta(\Omega))^2, \quad 1 < \beta \leqslant 2. \end{aligned}$$

Next we define the operators $\mathcal{A} : X \longrightarrow X'$ and $\mathcal{B} : X \longrightarrow V'$ as follows:

$$\langle \mathcal{A}(\sigma, p), (\tau, q)\rangle = \int_\Omega \Phi_\alpha(\sigma) : \tau \, \mathrm{d}x := \sum_{i,j=1}^{2} \int_\Omega |\sigma|^{\alpha-2}\sigma_{ij}\tau_{ij} \, \mathrm{d}x$$

$$\mathcal{B}(\sigma, p) = -\nabla \cdot (\sigma - pI)\cdot$$

$\langle \cdot, \cdot \rangle$ denotes the duality pairing between a Banach space and its dual space. Then (1.8) is formulated in the following variational form. Given $\mathbf{f} \in V'$

$$\begin{cases} \text{find } (\sigma, p) \in X \text{ and } \mathbf{u} \in V \text{ such that} \\ \langle \mathcal{A}(\sigma, p), (\tau, q)\rangle - \langle \mathcal{B}(\tau, q), \mathbf{u}\rangle = 0 \quad \forall (\tau, q) \in X, \\ \langle \mathcal{B}(\sigma, p), \mathbf{v}\rangle = \langle \mathbf{f}, \mathbf{v}\rangle \quad \forall \mathbf{v} \in V. \end{cases} \tag{2.5}$$

By use of Green's formula it may be shown that $(\sigma, p, \mathbf{u})$ such that $\sigma = \Phi_\alpha^{-1}(\nabla \mathbf{u})$ is a smooth solution of (2.5) if and only if $(\mathbf{u}, p)$ is a solution of (1.1) in the sense of distributions. We can also formulate problem (2.5) as

$$\begin{cases} \text{find } (\sigma, p) \in Z(\mathbf{f}) \text{ such that} \\ \langle \mathcal{A}(\sigma, p), (\tau, q)\rangle = 0 \quad \forall (\tau, q) \in Z(0). \end{cases} \tag{2.6}$$

In order to apply Theorem 1, we need to verify conditions (A1), (A2), (A3) and (B1). We begin with the following properties of the operator Φ_α.

PROPOSITION 2 Let $\Phi_\alpha(s) = |s|^{\alpha-2}s$, then there exist constants $M > 0$, $C > 0$ such that

(H1) $\langle \Phi_\alpha(s) - \Phi_\alpha(t), s - t\rangle \geqslant C\,|s-t|^\alpha,\ \forall s, t \in I\!R^4,$

(H2) $|\Phi_\alpha(s) - \Phi_\alpha(t)| \leqslant M\,(|s| + |t|)^{\alpha-2}|s-t|, \qquad \forall s, t \in I\!R^4.$

Proof. See Ciarlet (1978), Glowinski & Marroco (1975). □

REMARK 2 We also have the following properties of the the operator Φ_α (Barrett & Liu (1993, 1994), Sandri (1993)). There exist constants $M > 0$, $C > 0$ such that

$$\langle \Phi_\alpha(\sigma) - \Phi_\alpha(\tau), \sigma - \tau\rangle \geqslant C\left\{\|\sigma - \tau\|_{0,\alpha}^\alpha + \int_\Omega (|\sigma| + |\tau|)^{\alpha-2}|\sigma - \tau|^2\,\mathrm{d}x\right\},$$

$$\|\Phi_\alpha(\sigma) - \Phi_\alpha(\tau)\|_{0,\beta} \leqslant M\left[\int_\Omega (|\sigma| + |\tau|)^{\alpha-2}|\sigma - \tau|^2\,\mathrm{d}x\right]^{1/2} [\|\sigma\|_{0,\alpha} + \|\tau\|_{0,\alpha}]^{(\alpha-2)/2},$$

$\forall \sigma, \tau \in (L^\alpha(\Omega))^4$.

We now prove some auxiliary results that will be used in the proof of the existence and uniqueness theorem for problem (2.5).

LEMMA 1 Given $q \in L_0^r(\Omega)$, there exists $\mathbf{v} \in (W_0^{1,r}(\Omega))^2$ such that

$$\begin{aligned} \nabla \cdot \mathbf{v} &= q \\ \|\mathbf{v}\|_{1,r} &\leqslant C\|q\|_{0,r}, \quad \forall r > 1 \end{aligned} \tag{2.7}$$

where $C = C(r, \Omega)$.

Proof. See Galdi (1994). □

LEMMA 2 Let $1 < \beta \leqslant 2$ and $\alpha \geqslant 2$ be defined by $\frac{1}{\alpha} + \frac{1}{\beta} = 1$ and let $q \in L_0^\beta(\Omega)$. Then there exist $\mathbf{v} \in (W_0^{1,\alpha}(\Omega))^2$ and a constant $C > 0$ such that

$$\|\mathbf{v}\|_{1,\alpha,\Omega} \leqslant C\|q\|_{0,\beta,\Omega}^{\beta-1} \tag{2.8}$$

and we have

$$\int_\Omega q\nabla \cdot \mathbf{v}\,\mathrm{d}x = \|q\|_{0,\beta,\Omega}^\beta. \tag{2.9}$$

Proof. Let $q \in L_0^\beta(\Omega)$ and put $q_* = |q|^{\beta-2}q$ and $q_{**} = q_* - \frac{1}{|\Omega|}\int_\Omega q_*\,\mathrm{d}x$, where $|\Omega| = \int_\Omega \mathrm{d}x$. Then $q_{**} \in L_0^\alpha(\Omega)$ and from Lemma 2.1, there exist $\mathbf{v} \in (W_0^{1,\alpha}(\Omega))^2$ and a constant $C > 0$ such that

$$\nabla \cdot \mathbf{v} = q_{**}, \quad \text{and} \quad \|\mathbf{v}\|_{1,\alpha,\Omega} \leqslant C\|q_{**}\|_{0,\alpha,\Omega} \leqslant C\|q\|_{0,\beta,\Omega}^{\beta-1}$$

and we have

$$\int_\Omega q\nabla \cdot \mathbf{v}\,\mathrm{d}x = \int_\Omega qq_{**}\,\mathrm{d}x = \int_\Omega qq_*\,\mathrm{d}x = \|q\|_{0,\beta,\Omega}^\beta.$$

□

LEMMA 3 There exists a constant $C > 0$ such that, for every $\theta \in (L^\alpha(\Omega))^4$ satisfying $\nabla \cdot \theta = 0$ and $\int_\Omega \mathrm{tr}(\theta)\,\mathrm{d}x = 0$, we have

$$\|\theta\|_{0,\alpha} \leqslant C\|\theta - \tfrac{1}{2}\mathrm{tr}(\theta)I\|_{0,\alpha}. \tag{2.10}$$

Proof. Let $\theta^D = \theta - \frac{1}{2}\mathrm{tr}(\theta)I$. We shall show that

$$\|\mathrm{tr}(\theta)\|_{0,\alpha} \leqslant C\|\theta^D\|_{0,\alpha}.$$

By Lemma 1, there exits $\mathbf{v} \in W_0^{1,\beta}(\Omega)^2$ such that:

$$\nabla \cdot \mathbf{v} = q_* - \frac{1}{|\Omega|}\int_\Omega q_*\,\mathrm{d}x,\ q_* = |\mathrm{tr}(\theta)|^{\alpha-2}\mathrm{tr}(\theta) \in L^\beta(\Omega)$$

and

$$\|\mathbf{v}\|_{1,\beta} \leqslant C\|q_*\|_{0,\beta} = C\|\mathrm{tr}(\theta)\|_{0,\alpha}^{\alpha-1}.$$

Now

$$\begin{aligned}
\|\mathrm{tr}(\theta)\|_{0,\alpha}^\alpha &= \int_\Omega |\mathrm{tr}(\theta)|^\alpha\,\mathrm{d}x \\
&= \int_\Omega |\mathrm{tr}(\theta)|^{\alpha-2}\mathrm{tr}(\theta)\mathrm{tr}(\theta)\,\mathrm{d}x \\
&= \int_\Omega \nabla\cdot\mathbf{v}\,\mathrm{tr}(\theta)\,\mathrm{d}x \qquad \left(\text{since } \int_\Omega \mathrm{tr}(\theta)\,\mathrm{d}x = 0\right) \\
&= \langle\theta, \nabla\cdot\mathbf{v}I\rangle \\
&= 2\langle\theta, \nabla\mathbf{v} - \nabla\mathbf{v}^D\rangle \\
&= -2\langle\theta^D, \nabla\mathbf{v}\rangle \qquad (\text{since } \nabla\cdot\theta = 0) \\
&\leqslant 2C\|\theta^D\|_{0,\alpha}\|\mathrm{tr}(\theta)\|_{0,\alpha}^{\alpha-1}.
\end{aligned}$$

Hence

$$\|\mathrm{tr}(\theta)\|_{0,\alpha} \leqslant C\|\theta^D\|_{0,\alpha}$$

and $\|\theta\|_{0,\alpha} \leqslant C\|\theta^D\|_{0,\alpha}$. □

LEMMA 4 Let $Z(0) = \{(\tau, q) \in X, \quad \nabla\cdot(\tau - qI) = 0\}$. Then there exists a constant $C > 0$ such that

$$\|q\|_{0,\alpha} \leqslant C\|\tau\|_{0,\alpha}, \quad \forall(\tau, q) \in Z(0). \tag{2.11}$$

Proof. We generalize and we adopt the same proof as in Farhloul & Fortin (1993). Let $\theta = \tau - qI - rI$ where $r = \frac{1}{2|\Omega|}\int_\Omega \mathrm{tr}(\tau)\,\mathrm{d}x$. We have $\nabla\cdot\theta = 0$ and $\int_\Omega \mathrm{tr}(\theta)\,\mathrm{d}x = 0$. Also $\theta^D = \theta - \frac{1}{2}\mathrm{tr}(\theta)I = \tau - \frac{1}{2}\mathrm{tr}(\tau)I$. Then using Lemma 2.3 we get

$$\begin{aligned}
\|\tau - qI\|_{0,\alpha} &\leqslant \|\theta\|_{0,\alpha} + \|rI\|_{0,\alpha} \\
&\leqslant C\|\theta^D\|_{0,\alpha} + \|rI\|_{0,\alpha} \\
&\leqslant C\|\tau\|_{0,\alpha}.
\end{aligned}$$

This last inequality, with the fact that

$$\|q\|_{0,\alpha} \leqslant \|\tau - qI\|_{0,\alpha} + \|\tau\|_{0,\alpha},$$

gives us $\|q\|_{0,\alpha} \leqslant C\|\tau\|_{0,\alpha}$. □

PROPOSITION 3 (LBB condition for the velocity field) There exits a constant $C > 0$ such that

$$\sup_{(\tau,q)\in X} \frac{\langle \mathcal{B}(\tau, q), \mathbf{v}\rangle}{\|(\tau, q)\|_X} \geqslant C\,\|\mathbf{v}\|_V, \quad \forall \mathbf{v} \in V. \tag{2.12}$$

Proof. Let $\mathbf{v} \in V$ and let $\mathbf{w}$ denote the solution of the β-Laplacian problem

$$\begin{cases} -\nabla \cdot (|\nabla \mathbf{w}|^{\beta-2}\nabla \mathbf{w}) = |\mathbf{v}|^{\beta-2}\mathbf{v} & \text{in } \Omega \\ \mathbf{w} = 0 & \text{on } \Gamma. \end{cases}$$

From Ciarlet (1978), Glowinski & Marroco (1975), we know that this problem has a unique solution $\mathbf{w} \in (W_0^{1,\beta}(\Omega))^2$ and there exists a constant $C > 0$ such that

$$\|\mathbf{w}\|_{1,\beta} \leqslant C\|\mathbf{v}\|_{0,\beta}. \tag{2.13}$$

Therefore, $\theta = (|\nabla \mathbf{w}|^{\beta-2}\nabla \mathbf{w}, 0)$ is in the space X and there exists a constant $C > 0$ such that

$$\begin{aligned} \|(\theta, 0)\|_X^\alpha &= \|\theta\|_{0,\alpha}^\alpha + \|\nabla \cdot (\theta - 0I)\|_{0,\alpha}^\alpha \\ &= \int_\Omega |\nabla \mathbf{w}|^\beta \,\mathrm{d}x + \int_\Omega |\mathbf{v}|^\beta \,\mathrm{d}x \\ &\leqslant \|\mathbf{w}\|_{1,\beta}^\beta + \|\mathbf{v}\|_{0,\beta}^\beta \\ &\leqslant \max\,(1, C^\beta)\|\mathbf{v}\|_{0,\beta}^\beta. \end{aligned}$$

Hence

$$\|(\theta, 0)\|_X \leqslant \max\,(1, C^\beta)^{\frac{1}{\alpha}}\|\mathbf{v}\|_{0,\beta}^{\beta-1}$$

and then we have

$$\frac{\langle \mathcal{B}(\theta, 0), \mathbf{v}\rangle}{\|(\theta, 0)\|_X} \geqslant \frac{\|\mathbf{v}\|_{0,\beta}^\beta}{\max\,(1, C^\beta)^{\frac{1}{\alpha}}\|\mathbf{v}\|_{0,\beta}^{\beta-1}} = \frac{1}{\max\,(1, C^\beta)^{\frac{1}{\alpha}}}\|\mathbf{v}\|_{0,\beta}.$$

□

PROPOSITION 4 We have the following results:

(A1) $\mathcal{A}$ is hemicontinuous from X to X'.

(A2) There exists a constant $C > 0$ such that

$$\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\sigma, p) - (\tau, q)\rangle \geqslant C\|(\sigma, p) - (\tau, q)\|_X^\alpha, \quad \forall (\sigma, p), (\tau, q) \in Z(\mathbf{f}).$$

(A3) There exists a constant $M > 0$ such that

$$\|\mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q)\| \leqslant M(\|(\sigma, p)\|_X + \|(\tau, q)\|_X)^{\alpha-2}\|(\sigma, p) - (\tau, q)\|_X, \quad \forall (\sigma, p), (\tau, q) \in X.$$

Proof. Let $(\sigma, p), (\tau, q) \in Z(\mathbf{f})$. From $(H1)$ in Proposition 2.1 we have

$$\begin{aligned}\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\sigma, p) - (\tau, q)\rangle &= \int_\Omega (\Phi_\alpha(\sigma) - \Phi_\alpha(\tau)) : (\sigma - \tau)\, \mathrm{d}x \\ &\geqslant C \int_\Omega |\sigma - \tau|^\alpha \, \mathrm{d}x \\ &= C\|\sigma - \tau\|_{0,\alpha}^\alpha.\end{aligned}$$

Since $(\sigma - \tau, p - q) \in Z(0)$, the application of Lemma 2.4 yields

$$\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\sigma, p) - (\tau, q)\rangle \geqslant C\|(\sigma, p) - (\tau, q)\|_X^\alpha.$$

Now, for each $(\sigma, p), (\tau, q), (\theta, r) \in X$ we have

$$\begin{aligned}|\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\theta, r)\rangle| &= \left| \int_\Omega (\Phi_\alpha(\sigma) - \Phi_\alpha(\tau)) : \theta \, \mathrm{d}x \right| \\ &\leqslant \int_\Omega |(\Phi_\alpha(\sigma) - \Phi_\alpha(\tau))|\, |\theta| \, \mathrm{d}x\end{aligned}$$

and the identity

$$\|\mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q)\|_{X'} = \sup_{(\theta,r)\in X} \frac{\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\theta, r)\rangle}{\|(\theta, r)\|_X}.$$

The application of (H2) and the Holder inequality yield

$$|\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\theta, r)\rangle| \leqslant M\|(\sigma, p) - (\tau, q)\|_X\|(\theta, r)\|_X \left(\int_\Omega (|\sigma| + |\tau|)^\alpha \, \mathrm{d}x \right)^{\frac{\alpha-2}{\alpha}}.$$

We now use the triangle inequality to obtain the bound

$$|\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\theta, r)\rangle| \leqslant M\|(\sigma, p) - (\tau, q)\|_X\|(\theta, r)\|_X(\|\sigma\|_{0,\alpha} + \|\tau\|_{0,\alpha})^{\alpha-2}$$

and hence the result $(A3)$. □

REMARK 3 We observe that, from the definition of $\mathcal{A}$ and Remark 2.2, we also have the following inequalities:

$$\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q), (\sigma, p) - (\tau, q)\rangle \geqslant C \left\{ \|\sigma - \tau\|_{0,\alpha}^\alpha + \int_\Omega (|\sigma| + |\tau|)^{\alpha-2}|\sigma - \tau|^2 \, \mathrm{d}x \right\},$$

and

$$\|\mathcal{A}(\sigma, p) - \mathcal{A}(\tau, q)\|_{0,\beta} \leqslant M \left[\int_\Omega (|\sigma| + |\tau|)^{\alpha-2}|\sigma - \tau|^2 \, \mathrm{d}x \right]^{1/2} [\|\sigma\|_{0,\alpha} + \|\tau\|_{0,\alpha}]^{(\alpha-2)/2},$$

for all $(\sigma, p), (\tau, q) \in X$.

As a consequence of Theorem 2.1, Propositions 2.2 and 2.3, we have the following main result.

THEOREM 2 Problem (2.5) has a unique solution $((\sigma, p), \mathbf{u}) \in X \times V$ and there exists a constant $C > 0$ such that

$$\|(\sigma, p)\|_X + \|\mathbf{u}\|_V \leqslant C(\|\mathbf{f}\| + \|\mathbf{f}\|^{\alpha-1}).$$

3. Finite element approximations

We are looking for a discretization of (2.5) by triangular finite elements. Let $(\mathcal{T}_h)$ be a regular partition of Ω into triangles K. Let X_h and V_h be finite-dimensional subspaces of X and V. The discrete form of (2.5) is:

$$\begin{cases} \text{find } (\sigma_h, p_h) \in X_h \text{ and } \mathbf{u}_h \in V_h \text{ such that} \\ \langle \mathcal{A}(\sigma_h, p_h), (\tau_h, q_h)\rangle - \langle \mathcal{B}(\tau_h, q_h), \mathbf{u}_h\rangle = 0 \\ \langle \mathcal{B}(\sigma_h, p_h), \mathbf{v}_h\rangle = \langle \mathbf{f}, \mathbf{v}_h\rangle \\ \forall (\tau_h, q_h) \in X_h, \forall \mathbf{v}_h \in V_h. \end{cases} \tag{3.1}$$

We denote by $Z_h(\mathbf{f})$ the following space:

$$Z_h(\mathbf{f}) = \{(\tau_h, q_h) \in X_h,\ \langle \mathcal{B}(\tau_h, q_h), \mathbf{v}_h\rangle = \langle \mathbf{f}, \mathbf{v}_h\rangle,\ \forall \mathbf{v}_h \in V_h\}.$$

We will employ the following abstract results of Scheurer (1977) to study the finite-dimensional version of problem (2.5).

THEOREM 3 We assume that the following hypotheses hold:

($\mathrm{A2}_h$) There exists a constant $C_1 > 0$ independent of h such that, for $\alpha \geqslant 2$,

$$\begin{aligned} &\langle \mathcal{A}(\sigma_h, p_h) - \mathcal{A}(\tau_h, q_h), (\sigma_h, p_h) - (\tau_h, q_h)\rangle \\ &\geqslant C_1 \|(\sigma_h, p_h) - (\tau_h, q_h)\|_X^{\alpha},\ \forall (\sigma_h, p_h), (\tau_h, q_h) \in Z_h(\mathbf{f}). \end{aligned} \tag{3.2}$$

($\mathrm{A3}_h$) There exists a constant $C_2 > 0$ independent of h such that, for $\alpha \geqslant 2$,

$$\begin{aligned} &\|\mathcal{A}(\sigma_h, p_h) - \mathcal{A}(\tau_h, q_h)\|_{X'} \\ &\leqslant C_2(\|(\sigma_h, p_h)\|_X + \|(\tau_h, q_h)\|_X)^{\alpha-2} \\ &\times \|(\sigma_h, p_h) - (\tau_h, q_h)\|_X,\ \forall (\sigma_h, p_h), (\tau_h, q_h) \in X_h. \end{aligned} \tag{3.3}$$

($\mathrm{B1}_h$) There exists a constant $C_3 > 0$ independent of h such that

$$\sup_{(\tau_h, q_h) \in X_h} \frac{\langle \mathcal{B}(\tau_h, q_h), \mathbf{v}_h\rangle}{\|(\tau_h, q_h)\|_X} \geqslant C_3 \|\mathbf{v}_h\|_V, \quad \forall \mathbf{v}_h \in V_h. \tag{3.4}$$

Let $\mathbf{f} \in V'$. Then (3.1) has a unique solution $((\sigma_h, p_h), \mathbf{u}_h) \in X_h \times V_h$ which satisfies the following error estimate:

$$\begin{aligned} &\|(\sigma - \sigma_h, p - p_h)\|_X + \|\mathbf{u} - \mathbf{u}_h\|_V \\ &\leqslant C_1^* \|(\sigma - \tau_h, p - q_h)\|_X^{\frac{1}{\alpha-1}} \\ &+ C_2^* \max\{\|\mathbf{u} - \mathbf{v}_h\|_V^{\frac{1}{\alpha-1}},\ |\mathbf{u} - \mathbf{v}_h\|_V\} \\ &\forall (\tau_h, q_h) \in X_h, \forall \mathbf{v}_h \in V_h. \end{aligned} \tag{3.5}$$

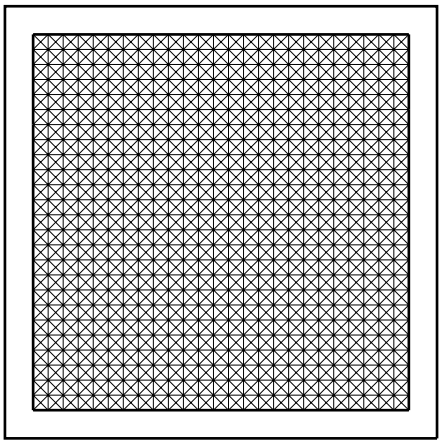


FIG. 1. Drive cavity mesh.

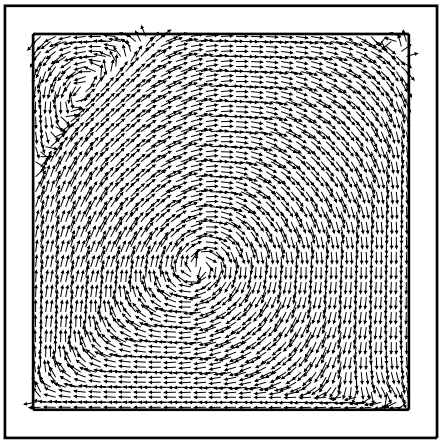


FIG. 2. Velocity vectors, $\beta = 1{\cdot}8$, $\quad \mathrm{Re} = \frac{1}{\nu_1} = 500$.

Proof. See Scheurer (1977). □

Now we turn to the finite element approximation of (3.1).

We introduce the space of the Raviart–Thomas elements (see Raviart & Thomas (1977), Nedelec (1980), Roberts & Thomas (1989)),

$$RT^k(K) = [P_k(K)]^2 \oplus \tilde{P}_k(K) \cdot (x, y)$$

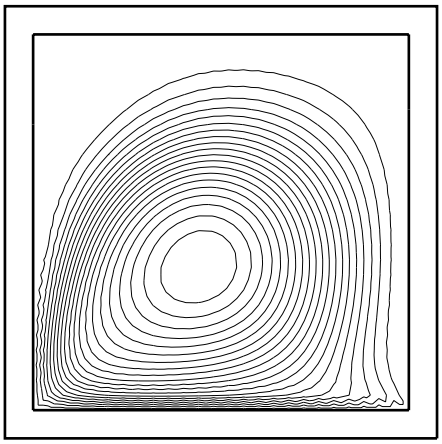


FIG. 3. Streamlines, $\beta = 1{\cdot}8$, $\quad \mathrm{Re} = \frac{1}{\nu_1} = 500.$

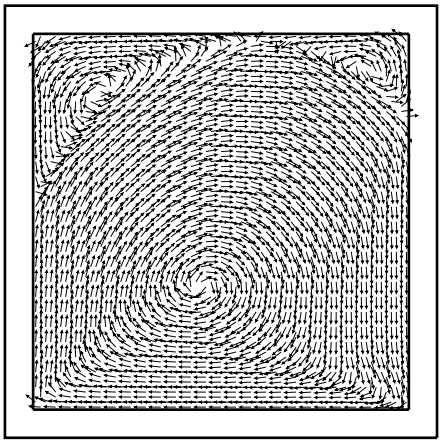


FIG. 4. Velocity vectors, $\beta = 3$, $\quad Re = \frac{1}{\nu_1} = 800.$

where $P_k(K)$ is the space of polynomials on K of degree at most k, $\tilde{P}_k(K)$ is the space of homogeneous polynomials of degree k. We make the following choice for X_h and V_h:

$$X_h = \{(\tau_h, q_h) \in X, \quad \tau_{h|K} \in (RT^k(K))^2, \quad q_{h|K} \in P_k(K), \forall K \in \mathcal{T}_h\}$$
$$V_h = \{\mathbf{v}_h \in V, \quad \mathbf{v}_{h|K} \in (P_k(K))^2, \forall K \in \mathcal{T}_h\}$$
$$Z_h(0) = \{(\tau_h, q_h) \in X_h; \quad \langle \mathcal{B}(\tau_h, q_h), \mathbf{v}_h \rangle = 0, \ \forall \mathbf{v}_h \in V_h\}.$$

Note that we have $Z_h(0) \subset Z(0)$. We also introduce the spaces

$$\Sigma_h = \{\tau_h \in (H^\alpha(\mathrm{div}, \Omega))^2, \quad \tau_{h|K} \in (RT^k(K))^2, \forall K \in \mathcal{T}_h\}$$
$$M_h = \{\mathbf{v}_h \in (L^\alpha(\Omega))^2, \quad \mathbf{v}_{h|K} \in (P_k(K))^2, \forall K \in \mathcal{T}_h\}.$$

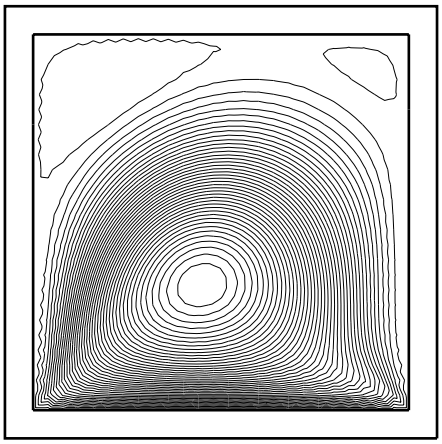


FIG. 5. Streamlines, $\beta = 3$, $\mathrm{Re} = \frac{1}{\nu_1} = 800$.

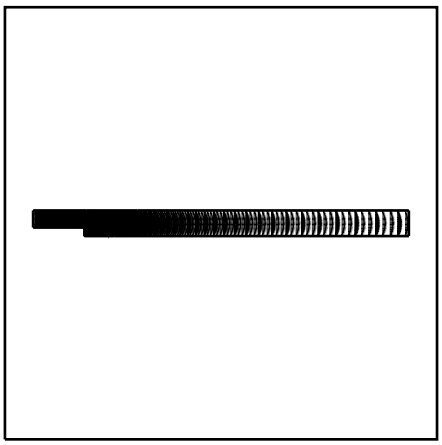


FIG. 6. Velocity vectors, $\beta = 1{\cdot}8$, $\mathrm{Re} = \frac{1}{\nu_1} = 50$.

Let $\Pi_h : (W^{1,\alpha}(\Omega))^4 \longrightarrow \Sigma_h$, and $P_h : (L^{\alpha}(\Omega))^2 \longrightarrow M_h$, the projection operators which have the following useful commuting property:

$$\mathrm{div} \circ \Pi_h = P_h \circ \mathrm{div} : (W^{1,\alpha}(\Omega))^4 \longrightarrow M_h$$

and the following approximation properties (Raviart & Thomas (1977), Milner (1985), Duran (1988)):

$$\|\mathbf{v} - P_h\mathbf{v}\|_{0,\alpha} \leqslant Ch^r \|\mathbf{v}\|_{r,\alpha}, \quad 0 \leqslant r \leqslant k+1, \quad 1 \leqslant \alpha \leqslant \infty, \tag{3.6}$$

$$\|\tau - \Pi_h\tau\|_{0,\alpha} \leqslant Ch^r \|\tau\|_{r,\alpha}, \quad \frac{1}{\alpha} < r \leqslant k+1, \quad 1 \leqslant \alpha \leqslant \infty, \tag{3.7}$$

$\forall \mathbf{v} \in (W^{r,\alpha}(\Omega))^2$ and $\forall \tau \in (W^{r,\alpha}(\Omega))^4$.

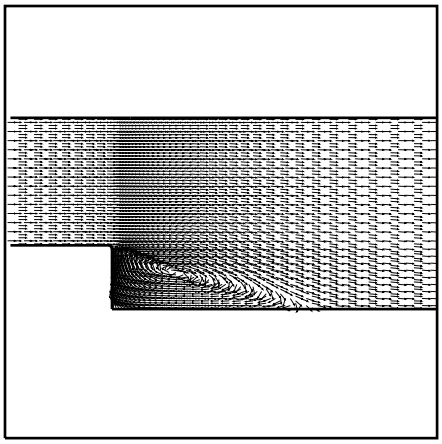


FIG. 7. Velocity vectors, $\beta = 1{\cdot}8$, $\quad \mathrm{Re} = \frac{1}{\nu_1} = 50$.



FIG. 8. Streamlines, $\beta = 1{\cdot}8$, $\quad \mathrm{Re} = \frac{1}{\nu_1} = 50$.

Before giving the error estimates for problem (3.1), let us prove the following result concerning the discrete inf-sup condition.

PROPOSITION 5 There exists a constant $C > 0$ independent of h such that

$$\sup_{(\tau_h, q_h) \in X_h} \frac{\langle \mathcal{B}(\tau_h, q_h), \mathbf{v}_h \rangle}{\|(\tau_h, q_h)\|_X} \geqslant C \, \|\mathbf{v}_h\|_V, \quad \forall \mathbf{v}_h \in V_h. \tag{3.8}$$

Proof. We suppose that $\alpha > 2$. In the case of $\alpha = 2$, this result is well known (see, Raviart & Thomas (1977), Girault & Raviart (1986)). Let $\mathbf{v}_h \in V_h$ and $\theta = |\nabla \mathbf{w}|^{\beta-2} \nabla \mathbf{w}$ where $\mathbf{w}$

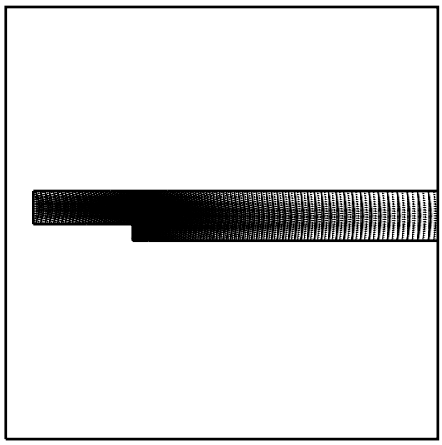


FIG. 9. Velocity vectors, $\beta = 3$, $\mathrm{Re} = \frac{1}{\nu_1} = 250$.

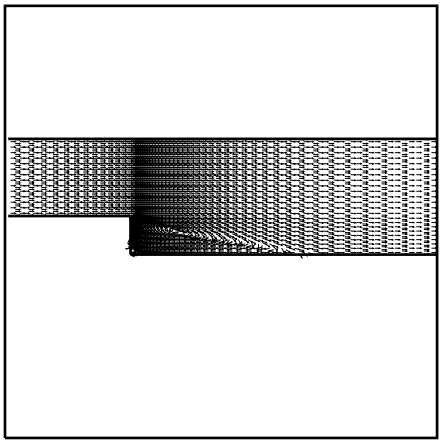


FIG. 10. Velocity vectors, $\beta = 3$, $\mathrm{Re} = \frac{1}{\nu_1} = 250$.

is the solution of the β-Laplacian problem

$$\begin{cases} -\nabla \cdot (|\nabla \mathbf{w}|^{\beta-2}\nabla \mathbf{w}) = |\mathbf{v}_h|^{\beta-2}\mathbf{v}_h & \text{in } \Omega, \\ \mathbf{w} = 0 & \text{on } \Gamma. \end{cases}$$

This problem has a unique solution $\mathbf{w} \in (W_0^{1,\beta}(\Omega))^2$ and there exists a constant $C > 0$ such that

$$\|\mathbf{w}\|_{1,\beta} \leqslant C\|\mathbf{v}_h\|_{0,\beta}. \tag{3.9}$$

Now, let $\tau_h^* = \Pi_h\theta$. Therefore, $(\tau_h^*, 0)$ is in the space X_h and there exists a constant

$C^* > 0$ such that (see the proof of Lemma 3.2 in Farhloul (1998))

$$\|(\tau_h^*, 0)\|_X = \|\Pi_h\theta\|_{0,\alpha} + \|\nabla \cdot \Pi_h\theta\|_{0,\alpha} \leqslant C^* \{\|\theta\|_{0,\alpha} + \|\nabla \cdot \theta\|_{0,\alpha}\}.$$

Hence

$$\|(\tau_h^*, 0)\|_X^\alpha \leqslant C_1 \{\|\mathbf{w}\|_{1,\beta}^\beta + \|\mathbf{v}_h\|_{0,\beta}^\beta\} \leqslant C_2 \|\mathbf{v}_h\|_{0,\beta}^\beta$$

which implies

$$\|(\tau_h^*, 0)\|_X \leqslant C_3 \|\mathbf{v}_h\|_{0,\beta}^{\beta-1}$$

and then

$$\frac{\langle \mathcal{B}(\tau_h^*, 0), \mathbf{v}_h\rangle}{\|(\tau_h^*, 0)\|_X} = \frac{\|\mathbf{v}_h\|_{0,\beta}^\beta}{\|(\tau_h^*, 0)\|_X} \geqslant C \|\mathbf{v}_h\|_{0,\beta}.$$

□

We are now in a position to state the error estimates for problem (3.1).

THEOREM 4 Problem (3.1) has a unique solution $((\sigma_h, p_h), \mathbf{u}_h) \in X_h \times V_h$. Further, if the solution $((\sigma, p), \mathbf{u})$ of problem (S_3) satisfies $((\sigma, p), \mathbf{u}) \in (W^{r,\alpha}(\Omega))^4 \times W^{r,\alpha}(\Omega) \times (W^{r,\beta}(\Omega))^2$, $1 \leqslant r \leqslant k+1$, $k \geqslant 0$, then there exists a constant C, independent of h, such that

$$\|\sigma - \sigma_h\|_{0,\alpha,\Omega} + \|p - p_h\|_{0,\alpha,\Omega} \leqslant C h^{2r/\alpha}, \tag{3.10}$$

$$\|\mathbf{u} - \mathbf{u}_h\|_{0,\beta,\Omega} \leqslant C h^r. \tag{3.11}$$

Proof. First we observe that the hypothesis $(A2_h)$ of Theorem 3.1 is a consequence of $(A2)$ of Proposition 2.3 and the fact that $Z_h(0) \subset Z(0)$. The hypothesis $(A3_h)$ is a direct consequence of $(A3)$, while $(B1_h)$ is proven in Proposition 3.1. Therefore, as a consequence of Theorem 3.1, problem (3.1) has a unique solution. To prove the error estimate for σ, first we observe that, from the second equation of (2.5) and of (3.1) respectively and the properties of the operator Π_h,

$$\langle \nabla \cdot (\Pi_h(\sigma - pI) - (\sigma_h - p_h I)), \mathbf{v}_h\rangle = 0, \quad \forall \mathbf{v}_h \in V_h. \tag{3.12}$$

Now let

$$(\sigma_h^*, p_h^*) = (\Pi_h(\sigma - pI) + s_h pI - \sigma_h, s_h p - p_h),$$

where, for all $q \in L^\alpha(\Omega)$,

$$s_h q = \rho_h q - \frac{1}{|\Omega|} \int_\Omega \rho_h q \, \mathrm{d}x$$

with ρ_h the projection operator from $L^\alpha(\Omega)$ onto $\prod_{K \in \mathcal{T}_h} P_k(K)$. Then, by using the first equation of (2.5) and of (3.1) respectively and the relations above, we get

$$\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h), (\sigma_h^*, p_h^*)\rangle = 0,$$

and then

$$\begin{aligned} &\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h), (\sigma, p) - (\sigma_h, p_h)\rangle \\ &= \langle \mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h), (\sigma, p) - (\Pi_h(\sigma - pI) + s_h pI, s_h p)\rangle. \end{aligned} \tag{3.13}$$

Now, using the fact that (cf. Remark 2.3)

$$\begin{aligned}&\langle \mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h), (\sigma, p) - (\sigma_h, p_h)\rangle \\ &\geqslant C \left\{ \|\sigma - \sigma_h\|_{0,\alpha}^{\alpha} + \int_{\Omega} (|\sigma| + |\sigma_h|)^{\alpha-2} |\sigma - \sigma_h|^2 \, \mathrm{d}x \right\}\end{aligned} \tag{3.14}$$

$$\begin{aligned}&\|\mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h)\|_{0,\beta} \\ &\leqslant C \left[\int_{\Omega} (|\sigma| + |\sigma_h|)^{\alpha-2} |\sigma - \sigma_h|^2 \, \mathrm{d}x \right]^{1/2} \\ &\times [\|\sigma\|_{0,\alpha} + \|\sigma_h\|_{0,\alpha}]^{(\alpha-2)/2}\end{aligned} \tag{3.15}$$

and Young's inequality, we get, for all $\epsilon > 0$,

$$\begin{aligned}&\|\sigma - \sigma_h\|_{0,\alpha}^{\alpha} + \int_{\Omega} (|\sigma| + |\sigma_h|)^{\alpha-2} |\sigma - \sigma_h|^2 \, \mathrm{d}x \\ &\leqslant \epsilon \int_{\Omega} (|\sigma| + |\sigma_h|)^{\alpha-2} |\sigma - \sigma_h|^2 \, \mathrm{d}x + \frac{C}{\epsilon} [\|\sigma\|_{0,\alpha} + \|\sigma_h\|_{0,\alpha}]^{\alpha-2} \\ &\times \|(\sigma, p) - (\Pi_h(\sigma - pI) + s_h p I, s_h p)\|_{0,\alpha}^2,\end{aligned} \tag{3.16}$$

where C is a constant independent of h and ϵ. On the other hand, owing to the inf-sup condition (3.8), we have $\|\sigma_h\|_{0,\alpha} \leqslant C$ with C a constant independent of h. Thus, (3.16) gives us

$$\begin{aligned}&ll \|\sigma - \sigma_h\|_{0,\alpha}^{\alpha} + \int_{\Omega} (|\sigma| + |\sigma_h|)^{\alpha-2} |\sigma - \sigma_h|^2 \, \mathrm{d}x \\ &\leqslant C \left(\|(\sigma - pI) - \Pi_h(\sigma - pI)\|_{0,\alpha} + \|p - s_h p\|_{0,\alpha} \right)^2.\end{aligned} \tag{3.17}$$

Therefore, the error estimate for σ is a consequence of (3.7), (3.17) and the following error estimate on the operator s_h:

$$\|p - s_h p\|_{0,\alpha} \leqslant C \, h^r \|p\|_{r,\alpha}. \tag{3.18}$$

To prove the error estimate for p, we observe that, from (3.12), $(\sigma_h^*, p_h^*) \in Z_h(0)$. Then, as in the proof of Lemma 2.4, we have $\|p_h^*\|_{0,\alpha} \leqslant C \|\sigma_h^*\|_{0,\alpha}$, where C is a constant independent of h. Thus,

$$\begin{aligned}\|s_h p - p_h\|_{0,\alpha} &\leqslant C \, \|\Pi_h(\sigma - pI) + s_h p I - \sigma_h\|_{0,\alpha} \\ &\leqslant C \, \{\|\Pi_h(\sigma - pI) - (\sigma - pI)\|_{0,\alpha} + \|\sigma - \sigma_h\|_{0,\alpha} + \|s_h p - p\|_{0,\alpha}\}.\end{aligned}$$

Therefore, the error estimate for p is a consequence of this last inequality, (3.7), the error estimate for σ, (3.18), and the triangle inequality. It remains to prove the error estimate for $\mathbf{u}$. First we observe that, from the first equation of (2.5) and of (3.1) respectively, we have

$$\langle \mathcal{B}(\tau_h, q_h), P_h \mathbf{u} - \mathbf{u}_h \rangle = \langle \mathcal{A}(\sigma, p) - \mathcal{A}(\sigma_h, p_h), (\tau_h, q_h) \rangle \quad \forall (\tau_h, q_h) \in X_h,$$

where $P_h\mathbf{u}$ is the projection of $\mathbf{u}$ onto $\left(\prod_{K\in\mathcal{T}_h} P_k(K)\right)^2$. Thus, using this last relation and the inf-sup condition (3.8), we get

$$C\|P_h\mathbf{u}-\mathbf{u}_h\|_{0,\beta} \leqslant \sup_{(\tau_h,q_h)\in X_h} \frac{\langle \mathcal{B}(\tau_h,q_h),\, P_h\mathbf{u}-\mathbf{u}_h\rangle}{\|(\tau_h,q_h)\|_X}$$

$$\leqslant \sup_{(\tau_h,q_h)\in X_h} \frac{\langle \mathcal{A}(\sigma,p)-\mathcal{A}(\sigma_h,p_h),(\tau_h,q_h)\rangle}{\|(\tau_h,q_h)\|_X}$$

$$\leqslant \|\mathcal{A}(\sigma,p)-\mathcal{A}(\sigma_h,p_h)\|_{0,\beta}.$$

Therefore, using (3.15) and (3.17), we obtain

$$\|P_h\mathbf{u}-\mathbf{u}_h\|_{0,\beta} \leqslant C\,\left(\|(\sigma-pI)-\Pi_h(\sigma-pI)\|_{0,\alpha}+\|p-s_hp\|_{0,\alpha}\right).$$

Finally, (3.11) is a consequence of this last inequality, (3.6), (3.7), (3.18) and the triangle inequality. □

REMARK 4 The error estimate (3.10) for $(\sigma,\, p)$ is better than the one obtained in Baranger *et al.* (1993), and the error estimate (3.11) for $\mathbf{u}$ is optimal.

4. Numerical results

Before giving some numerical results, we briefly present the iterative scheme for solving the system of non-linear equations resulting from (3.1) with $\nu_0 = 0$ and $\nu_1 > 0$ (see Ladyzhenskaya law defined by (1.2)). Subsequently, we drop the use of superscript h to denote discrete functions. We suppose that σ^0 is given such that $(\sigma^0, 0) \in X_h$; we define a Picard-type scheme for solving (S_h) as follows:

$$\begin{cases} \text{Find a sequence } [(\sigma^{m+1},\, p^{m+1}),\, \mathbf{u}^{m+1}] \in X_h \times V_h \text{ such that} \\ \frac{1}{\nu_1^{\alpha-1}} \int_\Omega |\sigma^m|^{\alpha-2}\sigma^{m+1} : \tau \,\mathrm{d}x + \int_\Omega \nabla\cdot(\tau - qI)\cdot\mathbf{u}^{m+1}\,\mathrm{d}x = 0, \quad \forall(\tau,q)\in X_h, \\ -\int_\Omega \nabla\cdot(\sigma^{m+1}-p^{m+1}I)\cdot\mathbf{v}\,\mathrm{d}x = \int_\Omega \mathbf{f}\cdot\mathbf{v}\,\mathrm{d}x, \quad \forall\mathbf{v}\in V_h. \end{cases}$$

For the numerical implementation of the above method, we introduce Lagrange multipliers λ_n^{m+1} and λ_t^{m+1} to relax the continuity of the normal trace $(\sigma^{m+1}-p^{m+1}I)\mathbf{n}$ across inter-element edges of the triangulation $\mathcal{T}_h$. These Lagrange multipliers λ_n^{m+1} and λ_t^{m+1} approximate the normal component $\mathbf{u}^{m+1}\cdot\mathbf{n}$ and the tangential component $\mathbf{u}^{m+1}\cdot\mathbf{t}$ of the velocity, respectively. The finite- dimensional space for the Lagrange multipliers is

$$\Lambda_h = \left\{(\mu_n,\mu_t) = (\mu\cdot\mathbf{n},\,\mu\cdot\mathbf{t}) \mid \mu_{|_e} \in (P_k(e))^2,\ \forall e\in\Gamma_h^0,\ \mu_{|_e}=0,\ \forall e\in\Gamma_h^\partial\right\}$$

with Γ_h being the set of edges of the elements of $\mathcal{T}_h$ and

$$\Gamma_h^\partial = \{e\in\Gamma_h,\ e\subset\Gamma\},\quad \Gamma_h^0 = \Gamma_h - \Gamma_h^\partial.$$

This procedure leads to a mixed-hybrid formulation for approximating the solution of (3.1), which is similar to the one in Farhloul & Fortin (1993). Hence, we use the same treatment

of the algebraic equations generated by the mixed-hybrid formulation as in Farhloul & Fortin (1993). At each step of the Picard-type scheme, the linear algebraic system which needs to be solved involves only the Lagrange multipliers and the pressure (see Farhloul & Fortin (1993) for more details). Finally, let us mention that, for the numerical results below, we have considered the problem (1.8) with the non-linear convection term $(\mathbf{u} \cdot \nabla)\mathbf{u}$, i.e. the first equation of (1.8) is replaced by the following one:

$$-\nabla \cdot (\sigma - pI) + (\mathbf{u} \cdot \nabla)\mathbf{u} = \mathbf{f} \quad \text{in } \Omega.$$

The non-linear term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ is treated as in Farhloul & Fortin (1993). One can also treat this term as in Farhloul & Manouzi (1996). The implementation is made with the lowest degree finite element, i.e. $k = 0$ for the spaces X_h, V_h and Λ_h. In this case, the Lagrange multipliers are constants on each edge of the triangulation and the pressure is constant on each triangle. We now present some numerical results on two test problems with the two values 1.8 and 3 for β (β being defined as $1/\alpha + 1/\beta = 1$). The first test is the driven cavity flow. The depicted domain is the square cavity $[0, 1] \times [0, 1]$, and the boundary conditions are $\mathbf{u} \cdot \mathbf{t} = 1$ and $\mathbf{u} \cdot \mathbf{n} = 0$ at $x = 0$, and $\mathbf{u} \cdot \mathbf{t} = \mathbf{u} \cdot \mathbf{n} = 0$ on the other parts of the boundary. The numerical results for this test problem are presented in Figs 2–5. The second test problem is the flow over a backward-facing step. The depicted domain is as follows: height 1, length before the step 3, height of the step 1/2, length after the step $L > 0$. While the boundary conditions are given as: at the inlet the profile of the velocity is parabolic, at the outlet $\mathbf{u} \cdot \mathbf{t} = 0$, and on the other parts of the boundary $\mathbf{u} \cdot \mathbf{t} = \mathbf{u} \cdot \mathbf{n} = 0$. The numerical results for this test problem are presented in Figs 6–10.

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Annex

We give here a correct proof of Lemma 1.3 of Scheurer (1977) without assuming that $\mathrm{Ker}(\mathcal{B})$ has a complement supplementary set as suggested by our anonymous referee. Let V and Y be two reflexif Banach spaces with norms $\|\cdot\|_V$ and $\|\cdot\|_Y$, respectively. Let V' and Y' be their dual spaces. We introduce a linear operator $\mathcal{B} \in \mathcal{L}(V, Y)$ and we denote by $\mathcal{B}' \in \mathcal{L}(Y', V')$ its adjoint operator, and by $\gamma(\mathcal{B})$ the reduced minimum modulus of $\mathcal{B}$ as defined in Kato (1976, p. 231). Then we have the following result.

LEMMA A.1 Let (a), (b) and (c) represent the following statements:

(a) There exits a constant $k > 0$ such that the compatibility condition

$$\|B'p'\|_{V'} \geqslant k\|p'\|_{Y'}, \ \forall p' \in Y', \tag{A.1}$$

holds.

(b) The operator $\mathcal{B}$ is surjective and

$$\gamma(\mathcal{B}) \geqslant k\cdot \tag{A.2}$$

(c) For all $g \in Y$ there exits $w \in V$ such that

$$\begin{cases} \mathcal{B}w = g \\ \|w\|_V \leqslant k^{-1}\|g\|_Y. \end{cases} \tag{A.3}$$

Then we have the following properties:

$$(a) \Longleftrightarrow (b) \Longrightarrow (c).$$

Proof. Let $N = \mathrm{Ker}(\mathcal{B})$ and $\tilde{V} = V/N$, the quotient space which is a Banach space with the norm defined by

$$\|\tilde{v}\|_{\tilde{V}} = \inf_{v \in \tilde{v}} \|v\|_V, \ \ \tilde{v} \in \tilde{V}.$$

We have the following results:

(i) If $\mathcal{B}'$ is injective then

$$\gamma(\mathcal{B}') = \inf_{p' \in Y',\, p' \neq 0} \frac{\|\mathcal{B}'p'\|}{\|p'\|} \qquad \text{(see Kato (1976))}\cdot$$

(ii)

$$\|\mathcal{B}v\| \geqslant \gamma(\mathcal{B})\|\tilde{v}\|, \ \forall v \in V \qquad \text{(see Kato (1976))}\ \cdot$$

(iii)

$$\gamma(\mathcal{B}) = \gamma(\mathcal{B}') \qquad \text{(see Kato (1976))}\ \cdot$$

(iv)

$$\mathcal{B} \text{ is surjective } \Longleftrightarrow \exists k > 0 \ \text{ such that } \ \|\mathcal{B}'p'\| \geqslant k\|p'\|, \ \forall p' \in Y'$$
(See Brezis (1992), Theorem. II-19 p. 29).

We then have the following relations:

$$\exists k > 0 \ \text{ such that } \ \|\mathcal{B}'p'\| \geqslant k\|p'\|, \ \forall p' \in Y'$$

$$\Longleftrightarrow \mathcal{B}' \ \text{ is injective and we have } \ \gamma(\mathcal{B}') \geqslant k \qquad \text{(by (i))}$$

$$\Longleftrightarrow \mathcal{B} \ \text{ is surjective and we have } \ \gamma(\mathcal{B}) \geqslant k \qquad \text{(by (iii))}$$

$$\Longrightarrow \forall g \in Y, \ \exists v \in V, \ \text{ such that } \ \mathcal{B}v = g, \ \text{ and } \ \|\tilde{v}\| \leqslant k^{-1}\|g\| \qquad \text{(by (ii) and (iv))}\cdot$$

But, since V is reflexive and from Brezis (1992) (Remark 6, p. 38 and Corollary III.20, p. 46) we can find $w \in \tilde{v}$ such that $\|w\| = \|\tilde{v}\|$. Therefore, using the last property we have that

$$\mathcal{B}w = g \ \text{ and } \ \|w\| \leqslant k^{-1}\|g\|$$

and the proof of the Lemma A.1 is complete. □

REMARK A.1 If the space V is uniformly convex then w is unique.